

Faranak Rajabi

Ph.D. Candidate, Mechanical Engineering (Computational Science) | M.S., Computer Science | UC Santa Barbara

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Education

University of California, Santa Barbara

Jan. 2022 - Jan. 2027 (expected)

Ph.D., Mechanical Engineering, GPA 3.94/4.00

Santa Barbara, CA

Advisors: Dr. Frédéric Gibou (Editor-in-Chief, *J. Comput. Phys.*) and Dr. Jeff Moehlis (Chair, Mech. Eng.).

Coursework: Applied Dynamical Systems, Linear Systems Theory, Kalman & Adaptive Filtering, Numerical Methods for PDEs, Stochastic Processes, Advanced Matrix Computations.

University of California, Santa Barbara

Sep. 2023 - June 2026

M.S., Computer Science, GPA 4.00/4.00 (concurrent second degree, full tuition fellowship)

Santa Barbara, CA

Coursework: Data Structures, Runtime Systems, Distributed Systems, Advanced Optimization Theory.

Sharif University of Technology

Sep. 2016 - Oct. 2021

B.Sc., Aerospace Engineering

Tehran, Iran

Rank 184 of 162,879 (top 0.11%) on the Iranian National University Entrance Exam; full tuition scholarship.

Technical Skills

Languages	C++ (11/14/17/20), Python, CUDA, C, MATLAB, Bash
GPU & HPC	CUDA, Kokkos, MPI, OpenMP, PETSc, p4est (AMR), kernel profiling, scaling to 1000+ cores
Simulation	Rigid body and continuum dynamics, signed distance fields, collision and contact interface tracking, ODE/PDE time integration, stability and convergence analysis
Control & Algorithms	Optimal control, HJB and dynamic programming, linear systems, Kalman filtering, convex optimization, algorithm complexity, data structures
ML & Tools	PyTorch, JAX (differentiable programming), TensorFlow, Git, Linux, CMake, Docker, CI/CD

Research Experience

Graduate Research Assistant, Computational Applied Science Lab (CASL)

Jan. 2022 - Present

UC Santa Barbara

Santa Barbara, CA

GPU-Accelerated Simulation and Control Solver (CUDA / Kokkos)

2022 - Present

- Building a **GPU-accelerated 4D Hamilton-Jacobi-Bellman solver** in **CUDA and Kokkos** for high-dimensional nonlinear dynamics, targeting real-time control and decision systems.
- Writing backend-portable kernels that compile to CPU and GPU from a single source; tuning memory access and occupancy to raise throughput without loss of fidelity.
- Built the first numerical framework for **globally optimal control of stochastic oscillator systems**; showed that coupled oscillators exhibit amplitude collapse, a mechanism invisible to reduced-order models (IEEE CDC 2026).

CASL-HJX: Open-Source High-Performance Simulation Library (Lead Developer)

2023 - Present

- Lead author and principal developer of a parallel **C++** simulation library for deterministic and stochastic Hamilton-Jacobi dynamics, released with full documentation and published in *Computer Physics Communications* (2025).
- Engineered an **adaptive mesh refinement pipeline delivering 50× speedup and 98% cell reduction**, concentrating resolution only where gradients demand it; scales to thousands of MPI ranks.

High-Fidelity Simulation of Moving Geometry and Contact Interfaces

2024 - Present

- Represent evolving geometry with **signed distance fields (level-set methods)** on adaptive octree grids, interface motion driven by local physical flux; the representation used for collision queries in physics engines.
- Built a sharp finite-volume solver on these moving interfaces, **second-order accurate in both solution and gradient** in L^∞ , validated in 2D and 3D on geometry with non-smooth corners.

First-Author Publications and Recognition

- **5 first-author** papers among 8 publications: “CASL-HJX: Deterministic and Stochastic Hamilton-Jacobi Solvers,” *Comput. Phys. Commun.* (2025); “Beyond Phase Reduction: Amplitude Collapse in Optimal Control of Coupled Oscillators,” *IEEE CDC* (2026); “Advection-Diffusion-Reaction on Piecewise Smooth Interfaces,” *J. Comput. Phys.* (under review).
- **Invited peer reviewer**, 11 manuscripts across 4 venues, incl. *J. Comput. Phys.* (7) and IEEE CDC (2). **Awards:** UCSB Dissertation Fellowship (2026-27, campus-wide); Graduate Summer Fellowship, four consecutive years.

Teaching and Leadership

- Authored and open-sourced a **Modern C++ course**: 204 curated examples across 29 chapters covering C++11/14/17/20, MIT-licensed and used beyond UCSB.
- **Graduate Teaching Assistant**, UCSB (2022-Present): 15+ appointments, 6 courses including Dynamics and Scientific Computing, 1000+ students.

Available on-site full time (40 hrs/week) for Fall 2026 via CPT; eligible for 36 months of OPT upon graduation.